

Improving Disease Diagnosis via Automated Image Analysis of Biological Tissue

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Abstract

Microgels are nanoscopic, biologically relevant particles, and the ability to detect them and count them precisely, is essential to a variety of biomedical applications, particularly in microfluidic diagnostics. This thesis presents a lightweight, interpretable, and reproducible approach which can be used to detect and count these microgels in video data using a custom-trained convolutional neural network (CNN). The project was based on such three microscopic videos of microgels flowing in a microfluidic channel where one video was annotated manually while other two represented independent test datasets. A modular pipeline was developed involving frame extraction, data cleaning, bounding box annotation using Roboflow and CNN-based binary classification. The model produced a precision and F1 score of more than 95 percent. These findings prove that task-specific CNN, with careful preprocessing and mathematical reasoning, can accommodate biomedical image analysis automation, even in the cases with a limited amount of data and computing resources. The project sets a fine premise of the subsequent work, and with that, it opens the gates to other such tasks like temporal tracking and morphological measurement.

Keywords: *microgel detection, convolutional neural network, image analysis, object detection, biomedical diagnostics, microfluidics, counting algorithm*

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Chapter 1: Introduction

Improving Disease Diagnosis via Automated Analysis of Biological Tissue

Microscopic evaluation of biological tissue is a fundamental technique in clinical diagnosis. It enables the scientists to examine the structure of cells, detect abnormalities, and detect early signs of disease, particularly in cancer pathology, hematology, and infectious diseases. In traditional microscopy, the stained tissue sample is viewed under an optical light microscope and this process depends entirely on a human expert who interprets visual patterns manually. Even though this method is reliable but is both time-consuming and prone to inter-observer variability. In more active high-throughput set-ups, it implies that diagnostic workflows tend to be clogged.

Recent progress in computational and digital imaging has introduced the possibility of automating various stages of tissue analysis. Such techniques are faster and more reproducible and consistent. This thesis focuses on one such area of automation: detecting and counting microgel structures which serves as biological cell analogues in microscopy videos. Microgels are synthetic, gel-like particles often used to mimic cell-like behavior in controlled experimental systems due to their uniform size and optical properties (Shieh et al., 2021). Figure 1.1 shows what a microgel looks like in the videos used in this thesis.



Figure 1.1: A microgel from the video data

The dataset used comprises three videos showing small, oval-shaped like microgels moving right to left inside a narrow microfluidic channel. These microgels were generated using a microfluidic platform that employed a T-junction configuration. The continuous phase and dispersed phase flowed through PTFE tubing to meet at the T-junction, enabling droplet formation through hydrodynamic focusing. The droplets were subsequently observed and recorded using an optical microscope connected to a computer for high-resolution video capture. Figure 1.2 shows a schematic representation of the physical setup that was used.

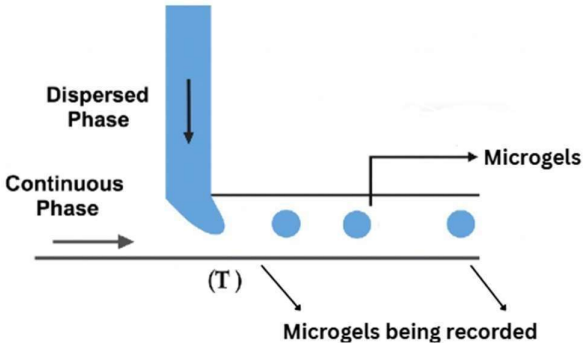


Figure 1.2: Schematic Diagram of Physical Setup

Figure 1.3 shows the experimental setup used for droplet generation and imaging. Panel A illustrates the T-junction connected to PTFE tubing, where the continuous phase and dispersed phase converge to produce microgels. Panel B depicts the integrated microfluidic platform, including the precision pump system, microscope, and computer interface used for visualizing and recording the droplet formation process.



Figure 1.3: The multifunctional integrated microfluidic platform. (A) T-junction connected to the PTFE tubing for two phases — continuous phase and dispersed phase — to generate droplets. (B) The microfluidic platform with the microscope for visualizing droplet formation

These microgels mimic the behavior of biological cells and are commonly used in laboratory studies as controlled substitutes. The aim of this project is to construct such end-to-end pipeline that can detect and count these microgels frame by frame.

To begin with, frames were extracted from the first video at 30 frames per second matching the original video's frame rate. Figure 1.4 shows some frames where a single microgel is passing by in the first video.

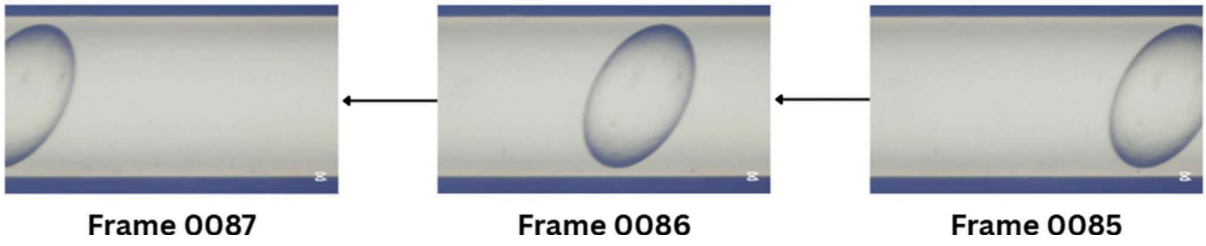


Figure 1.4: One microgel in different frames

These frames were then annotated using Roboflow, a web-based platform that facilitates image annotations and export to machine learning- ready training data formats (Wikipedia: Roboflow, 2025; Kale et al., 2024). Following annotation, a filtering logic is applied to remove poorly labelled frames, particularly those where the bounding box was incorrectly covering the entire image. Figure 1.5 shows two examples of bounding box annotation, where one image is poorly labelled and other image shows correct positioning of the bounding box.

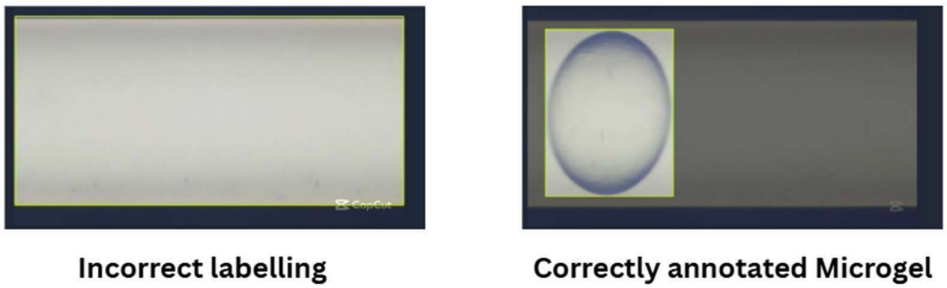


Figure 1.5: Incorrect and Correct labelling

Furthermore, the frames are classified into two categories, those with microgels and those without microgels. This de-cluttered dataset is hence used as the training data to train a simple Convolutional Neural Network (CNN) built from scratch. The trained model is subsequently tested on frames from the remaining two videos, and the number of microgels detected per video is recorded.

From mathematical point of view, the project gathers around basic geometry (e.g., bounding box coordinates), classification theory, and CNN architecture. Advanced statistical modelling is not the primary focus; however, efforts are made to ensure the model generalizes well to unseen data. All implementations are carried out using Python in Google Colab, emphasizing reproducibility and accessibility.

The broader importance of this work lies in its application in automation of biomedical diagnostics. The ability to detect and quantify cells or particles in microscopy videos is important for fields such as flow cytometry, cancer cell tracking, and microfluidic-based drug testing. Prior studies, such as Zarella et al. (2019), have shown that deep learning combined with digital pathology increases diagnostic

throughput, without negatively effecting accuracy. Similarly, Anagnostidis et al. (2020) showed real-time cell classification in microfluidic videos with neural networks as a bright example of how lightweight and versatile tools of computational powers are now used in laboratory diagnostics.

This thesis is structured as follows:

Chapter 2 presents a review of current literature, including object detection techniques, biological image analysis, and relevant computational frameworks.

Chapter 3 outlines the mathematical and computational methods used throughout the pipeline.

Chapter 4 details the experimental approach, including video processing, annotation, model training, and evaluation.

Chapter 5 presents the results and interprets model performance across test data.

Chapter 6 concludes the study, discusses its limitation, and proposes directions for future research.

By building a compact and functional detection pipeline, this study aims to demonstrate how foundational data science techniques can contribute to more scalable and reliable diagnostic systems in biomedical research.

Chapter 2: Literature Review

Automated tissue analysis has become a growing area of research within biomedical imaging, driven by the need for faster, more consistent, and scalable alternatives to manual microscopy (L. Pinto-Coelho et al., 2023; BMC Medical Imaging, 2024). The traditional diagnostics workflows still rely on human experts to visually inspect tissue slides or videos, but recent computational developments have made tools that can augment this human labor. The literature displays a broad diversity of methods from old-fashioned image processing to up-to-date neural networks, all employed in cell detection, object tracking, and quantitative tissue characterization.

One of the foundational areas in this field involves digital histopathology, where high-resolution images of tissue samples are processed algorithmically. Zarella et al. (2019) argued that reproducibility and interpretability are essential for any automated system entering clinical practice. As the paper emphasized, model transparency, the quality of its annotations, and its fit into the workflow of a health practitioner is as important as outright performance. Even the high-performing algorithms, medical trust and usability depend on reliable, consistent output. Figure 2.1 shows how a basic digital histopathology workflow looks.

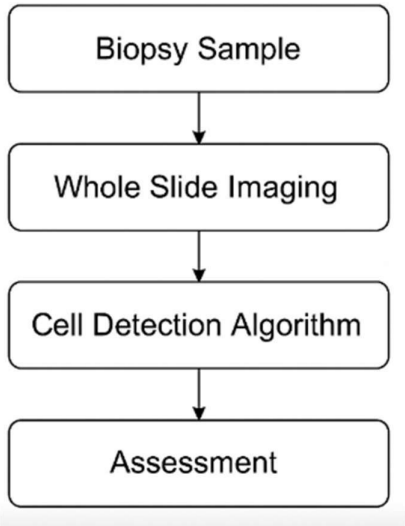


Figure 2.1: Digital Histopathology workflow

As research progressed, attention shifted towards analyzing live or dynamic tissue samples, often captured through microfluidic systems. This system is basically a miniature device designed to precisely control and manipulate small volumes of fluids, often at the scale of microliters and/or nanoliters. In microfluidic systems, fluids flow through networks of microchannels, enabling controlled interactions between different liquid streams. In the system used in this thesis, a T-junction configuration was employed, where one inlet carried the dispersed phase (cell-containing fluid) and another carried the continuous phase (oil or other carrier fluid). When these streams met at the junction, shear forces caused the dispersed phase to break into droplets, encapsulating the cells. These droplets, or microgels, were then carried downstream through the outlet channel, where their formation and movement were recorded via a microscope-mounted camera for subsequent analysis (as illustrated in Figure 1.2). It is important to have the video data prepared before any analysis by an algorithm. In many types of systems, biological material is transported through droplet-based microfluidic channels. They break down each individual cell into its own small droplet, making high-speed imaging in a controlled environment possible (Tang et al., 2024; Anagnostidis et al., 2020). For example,

brightfield microscopy allows images to be taken at high speed from videos at 1000-2000 fps. However, raw videos can have background noise, changes in lighting, and have air bubbles in it. The first step is preprocessing, which helps normalize contrast, eliminate anything static, and focus on the movement of the liquid. Frames are selected to help ease computational workload by sampling every nth frame. Tang et al. (2024) showed that carefully tuned preprocessing improved detection of cells by over 15%, especially when combined with morphological filters to get rid of debris. Technically, we ensure the data is standardized before detection. For example, automated pipelines often convert each frame to grayscale or normalize intensity, then it applies filter like Gaussian blur or morphological operations to reduce noise. Skipping frames can also make the process fast while maintaining the quality of detection, since cells do not move far between adjacent high-speed frames. Anagnostidis et al. (2020) demonstrated a platform where flowing cells were classified in real-time using compact image descriptors and neural networks. Their approach, like the present study, focused on high-speed frame-wise processing rather than deep architectural complexity. Similarly, Lin et al. (2023) explored how motion patterns and shape features could improve video-based cell classification, especially under conditions of noise or partial visibility.

To tackle detection tasks in such settings, many studies use object detection models that outputs bounding boxes around the items of interests. Mathematically, a bounding box is often represented as a tuple (x,y,w,h) where (x,y) indicates the box center or top-left corner, and (w,h) are the width and the height in pixel units. Figure 2.2 illustrates how a bounding box aligns on a x-y plot.

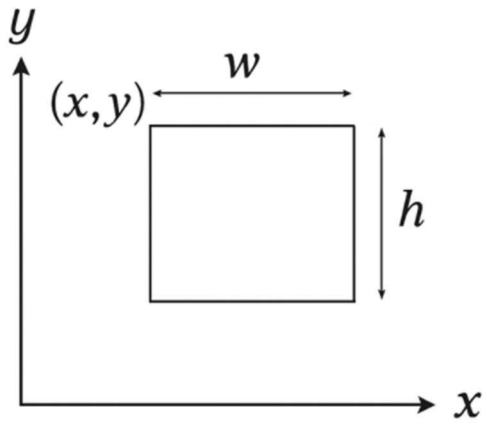


Figure 2.2: Bounding box on a x-y plot

These boxes allow algorithms to localize target objects within each frame. Some models like YOLO (You Only Look Once) or SSD (Single Shot MultiBox Detector) also assign a confidence score, which is a value in [0,1] indicating how likely it is that an object exists within the proposed region (Jocher et al., 2023) (Bochkovskiy et al., 2020). While Intersection over Union (IoU) is typically used to evaluate overlap between predicted and true bounding boxes (Bewley et al., 2016). In this thesis, it was used to calculate the precision and F1 score of the placement of the bounding box.
Figure 2.3 shows an example of how YOLO detects Red Blood Cells (RBCs) and White Blood Cells (WBCs).

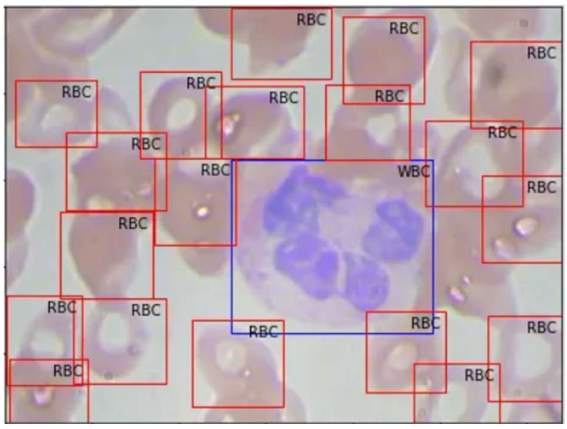


Figure 2.3: Blood Cell Detection using YOLOv3 (Source: https://medium.com/@admin_3546/blood-cells-detection-with-yolov3-darknet-d49ae63f79d9)

While many researchers apply large-scale pre-trained object detection models, they generally require high-levels of computational power to perform. On the contrary, task-specific solutions that are built on simple Convolutional Neural Networks can achieve comparable good results when tailored to well-defined tasks. Falk et al., (2019) showed that small CNNs trained on limited data could achieve strong performance in biomedical segmentation, which is basically segregating different biological matters such as tumors or cells or any area of interest, provided the data quality was high. The effectiveness of CNNs stems from their ability to detect spatial patterns by convolving filters across an image grid. Each filter extracts local features such as edges, textures, or contours, and successive layers build up more abstract representations.

After feeding the frame-wise image data into a CNN, the network ends up recognizing visual templates, which essentially approximate the usual shape and orientation of cells or cell-like structures. This is particularly useful in microfluidic videos where the specific direction of motion (e.g., consistently left to right) is at least predictable and the shape (e.g., ellipses or ovals) is relatively consistent.

In the area of cell counting, researchers have used both detection-based and density-based approaches. Detection-based methods localize each object instance typically with bounding boxes and count by enumerating detections while density-based methods predict a continuous “density map” whose integral gives the count without explicit boxes. Tiwari et al., (2023) presented a semi-supervised method to detect and count circulating tumor cells (CTCs) in sparse datasets. Their results suggested that even with minimal annotations, when combined with model fine-tuning (adapting a pre-trained model to the target dataset by continuing training) and confidence filtering (removing low-probability detections using a threshold to reduce false positives), it could yield reliable counts. However, their model assumed access to segmentation maps, which are pixel-wise labels that delineate the exact extent of each object and is more detailed than a bounding box, whereas the present study is limited to bounding-box annotations only.

The quality of annotation and data filtering are also deemed as important elements. According to Klie et al., (2022), noisy or inconsistent annotations can have disastrous effects on model performance, particularly in cases where object to background ratios are low, and in cases where classes are highly unbalanced. This understanding strengthens the choice in this thesis regarding the manual review of the annotations and the elimination of those frames where bounding box covers the whole frame which could cause an error that could mislead the model during training.

In terms of practical applications, automated video-based detection and counting systems are increasingly used in clinical and research environments, particularly in flow cytometry and microfluidic diagnostics. Tang et al., (2024) argued that lightweight, modular systems could serve as pre-screening tools, flagging relevant samples for further expert review. This aligns with the goal of the present study, which is not to build a diagnostic classifier but to develop a reusable counting tool that can reduce human effort and improve

reliability in biological video analysis.

Table 2.1 summarizes selected studies that exemplify the range of biomedical image analysis tasks relevant to this thesis. Zarella et al. (2019) demonstrated the use of deep learning in digital histopathology, emphasizing reproducibility and clinical integration. Anagnostidis et al. (2020) achieved real-time classification of cells in microfluidic video frames using compact feature-based neural networks. Falk et al. (2019) showed that small CNNs can perform effectively in biomedical segmentation tasks given high-quality labels. Tiwari et al. (2023) proposed a semi-supervised framework for counting circulating tumor cells in sparse microscopy videos, combining fine-tuning with confidence filtering. Klie et al. (2022) investigated annotation quality, finding strong links between annotation errors and drops in model accuracy. Finally, the present thesis develops a lightweight CNN-based pipeline for detecting and counting flowing microgels in microfluidic videos, with emphasis on annotation quality control and preprocessing tailored to the task. These comparisons also help conte contextualize the unique contribution of this study, which focuses on frame-wise object detection and counting in microfluidic video sequences using a lightweight CNN trained from scratch.

Author	Task	Data Type	Method Used	Application	Key Insight
Zarella et al., (2019)	Digital histopathology analysis	Static tissue images	Deep learning + pathologist review	Clinical pathology	Emphasized reproducibility and integration in clinical workflows
Anagnostidis et al., (2020)	Real-time cell classification	Microfluidic video frames	Feature-based NN	Flow-based single-cell diagnostics	Achieved real-time performance in classifying moving cells
Falk et al., (2019)	Biomedical segmentation	Cell microscopy images	U-Net CNN	General biomedical image tasks	Showed small CNNs can perform well with high-quality labels
Tiwari et al., (2023)	Counting circulating tumor cells	Sparse microscopy videos	Semi-supervised learning	Cancer diagnostics	Proposed a lightweight counting system under limited annotation availability
Klie et al., (2022)	Annotation quality assessment	Microscopy datasets	Statistical error analysis	Model accuracy studies	Demonstrated strong link between annotation quality and model performance
This thesis	Microgel detection and counting	Microfluidic video frames	CNN from scratch + preprocessing	Prototype for diagnostic support tools	Developed a lightweight pipeline for detecting and counting flowing microgels

Table 2.1: Overview of selected studies relevant to biomedical image analysis, with a focus on detection and counting tasks in video or microscopy data. The final row summarizes the present study's approach and positioning.

Overall, the reviewed literature supports the motivation and design of this thesis. It shows that targeted, carefully trained CNN models when combined with thoughtful annotation and cleaning strategies can achieve meaningful performance in biological object detection tasks. The next chapter is built on this background by introducing the mathematical and statistical methodology used throughout this project.

Chapter 3: Mathematical and Statistical Methodology

This chapter explains the mathematical concepts, computational models, and statistical evaluation methods that form the backbone of this study. It begins with a representation of digital images as numerical matrices, moving from grayscale to RGB formats, and discusses why such representations are essential for computer-based image analysis. The chapter then introduces neural networks and convolutional neural networks (CNNs), detailing their components, operations, and relevance to object detection in biomedical imaging. The specific CNN architecture implemented in this thesis is described, along with the custom loss function, optimization strategy, and regularization techniques used during training. Finally, statistical metrics for evaluating detection performance are presented, ensuring a clear link between theoretical foundations and the practical goal of detecting and counting microgels in microfluidic videos.

3.1 Digital Images as Matrices

Digital Images can be understood mathematically as matrices, arrays of numeric values where each element, called a pixel, represents brightness or colour intensity at that position. A typical grayscale digital image is represented by a two-dimensional matrix with pixel values ranging from 1 (black) to 0 (white) (Gonzalez and Woods, 2017). Such matrix representation allows the application of various mathematical and computational techniques directly to the images. Figure 3.1 shows how a grayscale image looks like a matrix.

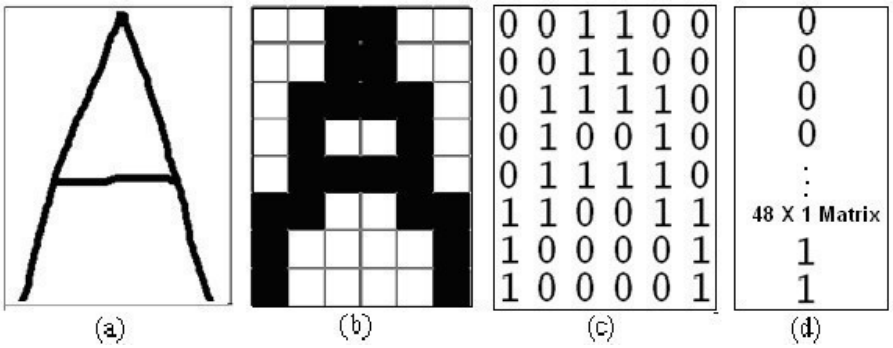


Figure 3.1: (a) Grayscale image of character 'A' (b) Binary representation of character 'A'; (c) Binary matrix representation and (d) Reshaped sample of character 'A' (Source: https://www.researchgate.net/figure/a-Grayscale-image-of-character-A-b-Binary-representation-of-character-A-c_fig6_50211012).

3.2 RGB Images versus Grayscale Images

Grayscale images contain one intensity value per pixel, represented by a single numeric matrix. In contrast, color images, such as RGB images, have three separate intensity values per pixel, corresponding to the red, green, and blue channels. Thus, RGB images can be viewed as three distinct matrices stacked together. Each pixel in an RGB image is effectively a three-dimensional vector, making these images richer in information but computationally more intensive compared to grayscale images (Szeliski, 2010). Often, RGB images are converted to grayscale to simplify analysis, especially for methods like edge detection.

Figure 3.2 shows a batch of 3 grayscale images that can be represented using a three-dimensional (3D) NumPy array or a tensor.

Similarly, figure 3.3 shows a batch of 3 RGB images that can be represented using a four-dimensional (4D) NumPy array or a tensor.

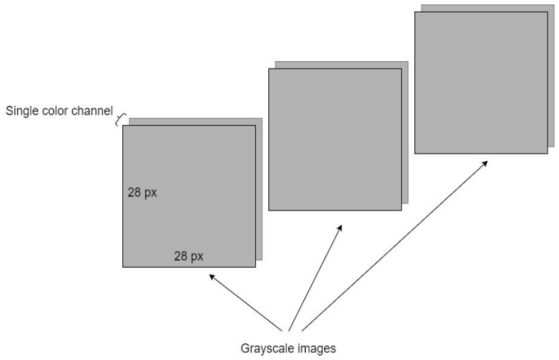


Figure 3.2: Grayscale Image Shape: (sample, height, width): (3,28,28) (Source: https://towardsdatascience.com/exploring-the-mnist-digits-dataset-7ff62631766a/)

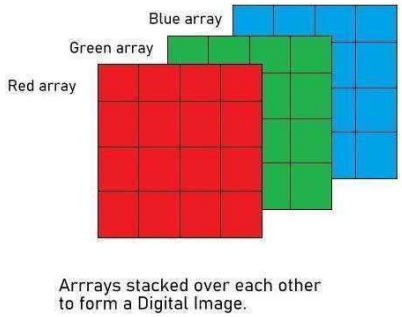


Figure 3.3: RGB Image Shape: (samples, height, width, color channel): (3,28,28,3) (Source: https://www.geeksforgeeks.org/software-engineering/matlab-rgb-image-representation/)

3.3 Neural Networks

Neural networks (NNs) are computational models inspired by biological neurons. They are powerful mathematical tools for capturing complex relationships within data, particularly useful for classification, regression, and pattern recognition problems.

Mathematical and Statistical Modelling

A neural network consists of many interconnected units called neurons (or perceptron). Each neuron takes input values, processes them using a set of weights and a bias term, and produces an output through an activation function. Mathematically, the output y of a neuron given input $x_1, x_2, \ldots, x_n$ can be described as:

$$y = activation\ (w_1x_1 + w_2x_2 + \cdots + w_nx_n + b)$$

Here:

- x_1 represents the input features
- w_1 represents weights assigned to each input, determining their relative importance
- b is a bias term, allowing adjustment of the activation threshold
- The activation function introduces non-linearity, allowing the network to learn complex patterns

Figure 3.4 illustrates how activation function work with neural networks.

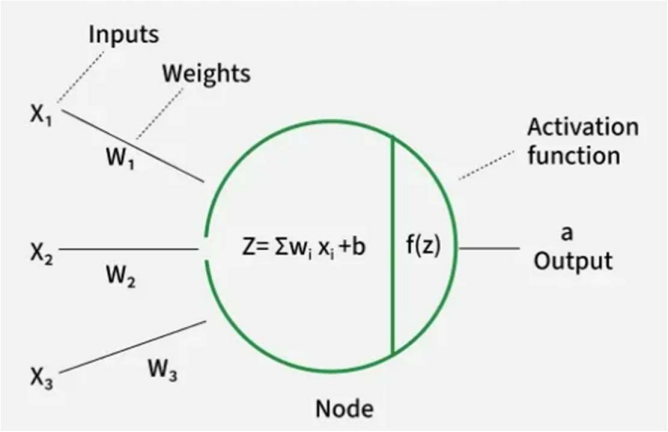


Figure 3.4: Activation Function in Neural Networks (Source: https://www.geeksforgeeks.org/machine-learning/activation-functions-neural-networks/)

3.3.1 Activation Functions

Activation functions are mathematical transformations applied to the output of a neuron before passing it to the next layer. Their purpose is to introduce non-linearity into the network, enabling it to learn complex relationships between inputs and outputs. Without them, the network would simply behave like a linear model, no matter how many layers it had.

We divide activation functions into internal (hidden layer) and output layer activations:

3.3.1.1. Internal (Hidden Layer) Activation Functions

- **Rectified Linear Unit (ReLU):** Very popular in modern neural networks because it is computationally efficient and focuses the network on active features while ignoring negative signals which speeds up the convergence. ReLU outputs the input directly if positive, otherwise zero.

$$ReLU(z) = \max(0, z)$$

- **Sigmoid function:** Maps input to a value between 0 and 1. Useful for binary classification problems. It gives a smooth, bounded output.

$$sigmoid\ (z) = \frac{1}{1 + e^{-z}}$$

- **Hyperbolic tangent (tanh):** Like sigmoid but maps inputs between -1 and 1, useful when data has positive and negative values. It helps with faster convergence when compared to sigmoid.

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

Figure 3.5 illustrates the graphs of these activation functions.

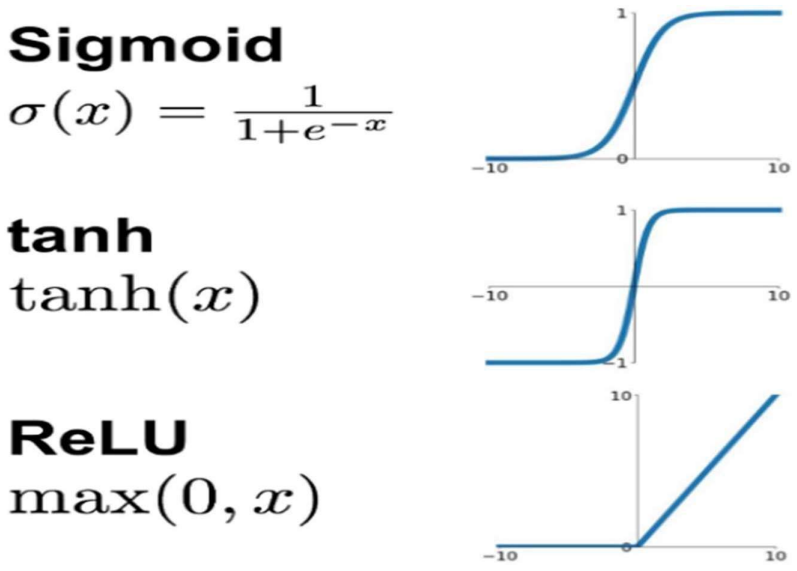


Figure 3.5: Activation Functions: a) Sigmoid b) tanh c) ReLU (Source: Mansir et al., 2022)

3.3.1.2. Output Layer Activation Functions

These shape the final network output to match the nature of the prediction task.

- **SoftMax:** It is used in multi-class classification and outputs probability distribution over K classes

$$softmax\ (z_i) = \frac{e^{z_i}}{\sum_{k=1}^{k} e^{z_k}}$$

- **Sigmoid function:** Maps input to a value between 0 and 1. Useful for binary classification and outputs probability of the positive class. It can be used in both hidden layer and output layer.
- **Linear function:** It is useful in regression tasks with unbounded continuous outputs.
- **ReLU/ ReLU6/ Softplus:** It is used in regression tasks where the target variable is non-negative and hence gives only positive outputs. ReLU6 is a variation of ReLU that caps the maximum output at 6, which can help stabilise training in certain architectures, particularly on mobile or quantised networks. Softplus is a smooth, differentiable approximation of ReLU that avoids sharp zero-cutoffs, allowing small gradients even for negative inputs (ReLU and Softplus activation n.d.).

To conclude, the hidden layer activations learn complex feature transformations while output layer activation ensures predictions respect the mathematical and physical constraints of the problem.

A single neuron processes one set of inputs to produce one output. By arranging neurons into layers, we can build a network capable of learning far more complex relationships. The input layer receives the raw features, the hidden layers perform successive transformations, and the output layer produces the final prediction. Each neuron in a hidden layer connects to every neuron in the next layer, passing its output forward. This structure enables the network to learn hierarchical representations, where early layers capture basic features (edges, textures) and deeper layers learn more abstract concepts (shapes, objects). In the context of this thesis, such a network takes in an image (or part of an image) and, through many layers of transformations, outputs either a probability of an object's presence, a set of class probabilities, or numeric predictions for bounding box coordinates. Figure 3.6 shows a typical feedforward neural network architecture with one input layer, two hidden layers, and one output layer.

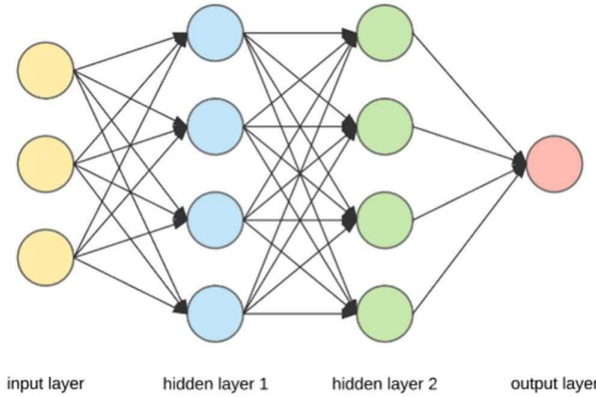


Figure 3.6: General Structure of a Neural Network (Source: https://www.geeksforgeeks.org/deep-learning/layers-in-artificial-neural-networks-ann/)

3.3.2 Loss Functions

Loss functions quantity how far the network's predictions are from the true target values and guide the optimization process by providing a signal for weight updates. The choice of loss functions must match both the output activation function and the nature of the prediction task. Some of the loss functions are:

a. Classification Loss (Cross-Entropy)

For classification problems, the goal is to compare the predicted probability distribution to the actual class labels.

- Binary Cross- Entropy (BCE) is used for binary classification with a sigmoid output. In this case each prediction $\hat{y}_i$ is a probability that that input belongs to the positive class:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

This penalizes confident but incorrect predictions heavily, encouraging the network to output accurate probabilities.

- Categorical Cross-Entropy (CCE) is the multi-class extension, used with a SoftMax output. It compares predicted and true probability distributions over multiple classes, penalizing divergence between them.

b. Continuous Regression Output Loss

When the network outputs a continuous value (e.g., microgel diameter, bounding box coordinates), the error is measured in the same units as the target.

- Mean Squared Error (MSE) penalizes larger deviations quadratically:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

- Root Mean Squared Error (RMSE) is the square root of MSE and is easier to interpret since it is in the same units as the target variable:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}$$

In this thesis, RMSE is used to evaluate regression accuracy for bounding box parameters (x,y,w,h).

c. Bounding Box Regression Loss

In object detection, the network must predict not just whether an object is present but also its position and size. The bounding box regression loss measures the difference between the predicted box coordinates (x, ŷ, ŵ, ĥ) and the ground-truth values. RMSE or MSE can be applied here, with the loss computed only for grid cells containing objects. This ensures the network focuses on refining box predictions for relevant regions.

In addition to coordinate regression, Intersection over Union (IoU) based losses directly evaluate spatial overlap between predicted and true boxes. IoU is defined as:

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

Higher IoU indicates better alignment. Variants such as GIoU (Generalized Intersection over Union), DIoU (Distance Intersection over Union), and CIoU (Complete Intersection over Union) introduce additional penalty terms to address shortcomings of the basic IoU metric. GIoU penalizes predictions that do not cover the ground truth box adequately even when there is no overlap, DIoU incorporates the normalized distance between box centers to improve convergence, and CIoU further accounts for aspect ratio consistency between predicted and ground truth boxes, improving robustness in detection tasks (Haurilet et al., 2019).

3.3.3 Optimization

During training, a neural network adjusts its internal parameters — the weights (w) and biases (b) — to minimise the difference between its predictions and the true labels. This difference is quantified using a loss function (or cost function), which serves as a measure of prediction error. The optimisation process seeks to find the set of parameters that minimises this loss.

The most widely used approach is gradient descent, where the network computes the gradient (partial derivative) of the loss function with respect to each parameter. The parameters are then updated in the opposite direction of the gradient, moving toward the minimum of the loss landscape:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}, b_{new} = b_{old} - \eta \frac{\partial L}{\partial b}$$

Where η (eta) is the learning rate determining the size of updates.

Here, η (eta) is the learning rate, which determines the step size of each update. A learning rate that is too high can cause instability, while a rate that is too low can result in very slow training. A crucial innovation that allows neural networks to scale to large datasets is Stochastic Gradient Descent (SGD), first formalized by Bottou (2010). Instead of computing the gradient over the entire dataset at once (which can be computationally expensive), SGD approximates the true gradient by computing it over small, randomly selected subsets of the data, called mini-batches. This introduces some noise into the updates, which can help escape shallow local minima and improve generalisation. Modern variants such as Adam (Adaptive Moment Estimation), introduced by Kingma and Ba (2015), combine the ideas of SGD with adaptive learning rates, accelerating convergence and improving stability.

The process of computing gradients and updating parameters in a network is known as backpropagation. Backpropagation efficiently applies the chain rule of calculus layer-by-layer from the output back to the input, ensuring that each weight and bias is updated according to its contribution to the final prediction error.

3.4 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are specialized neural networks tailored specifically for image and spatial data. Unlike traditional neural networks, CNNs explicitly leverage spatial relationships in images by employing convolution and pooling operations.

3.4.1 Convolutional Layer

Before CNNs, image processing often relied on manually designed filters to enhance specific image features or remove unwanted details. These filters are represented as small matrices or kernels that slide over the image, computing a weighted sum of pixel values within a local neighborhood. The output forms a new image called a feature map, where certain patterns (edges, textures, smooth regions) are enhanced.

For example, an averaging filter (also called a mean filter) assigns equal weights to each pixel in a neighborhood, replacing the central pixel with the average value. This smooths variations, resulting in a blurred image. The kernel for a 3 x 3 averaging filter is:

$$F_{avg} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

When applied, edges and fine details are reduced, while smooth regions remain largely unaffected.

In contrast, edge detection filters highlight areas of rapid intensity change, marking object boundaries. For instance, a Sobel filter in the horizontal direction can be written as:

$$F_{sobel-x} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Figure 3.7 shows how edges are emphasized by an application of an edge filter

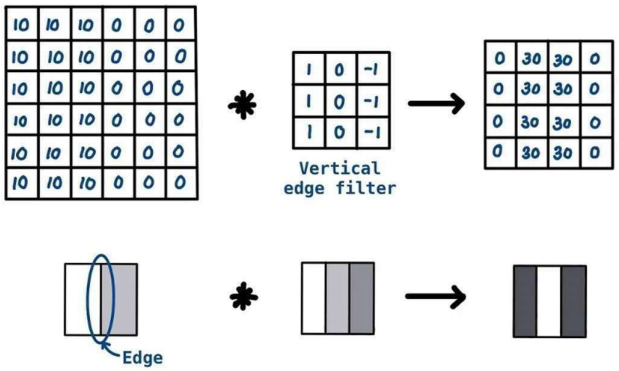


Figure 3.7: Edge Detection (Source: https://savindi-wijenayaka.medium.com/all-about-convolutional-neural-networks-cnns-172425c2cce4)

This kernel responds strongly to vertical edges by computing differences between pixel intensities in adjacent columns. Similarly, the Sobel-y kernel detects horizontal edges.

In classical image preprocessing, these filters are pre-specified by human designers to perform a known task- blurring, sharpening, or edge detection without any learning from data.

CNNs adopt the same sliding window principle but take it further. Instead of using fixed, hand-crafted filters, they learn the filter weights $F(u, v)$ automatically during training. This enables the network to discover complex, task-specific features directly from the data, from simple edges in early layers to intricate textures and shapes in deeper layers.

Formally, the convolution operation for a filter F of size m × m applied to an image I is given by:

$$C(i,j) = \sum_{u=1}^{m}\sum_{v=1}^{m} F(u,v).I(i+u-1, j+v-1)$$

Where:

- $C(i,j)$ is the resulting feature map at position (i,j)
- $F(u,v)$ are filter weights learned during training
- $I(i+u-1, j+v-1)$ is the pixel value in the input image at the corresponding position

Each filter captures a specific feature, and multiple filters create multiple feature maps. This approach significantly reduces the number of parameters compared to fully connected networks because filters are shared across all spatial positions.

3.4.2 Activation Function in CNNs

After convolution, an activation function (usually ReLU) is applied to introduce non-linearity to the feature maps. This step helps CNNs capture more complex patterns beyond linear features.

3.4.3 Pooling Layer

Pooling layers follow convolutional layers and reduce the spatial dimensions of feature maps, summarizing local regions. This reduction simplifies computation and helps CNNs recognize features regardless of minor spatial variations.

The most common pooling method is max pooling, which selects the maximum value within a small window (e.g., 2×2), effectively highlighting the most prominent features:

$$P(i,j) = \max\left(C_{\text{window}}\right)$$

where C_{window} represents all the values within the small local window of the feature map. Other methods like average pooling (taking the average value) are also used occasionally.

3.4.4 Fully Connected Layers and Output

After several convolutional and pooling layers, CNNs typically include one or more fully connected layers. These layers combine features learned by earlier layers to make final predictions. The final fully connected layer usually uses an activation function (like SoftMax) to produce probabilities for different classes.

The SoftMax function for multi-class classification is given by:

$$softmax(z_i) = \frac{e^{z_i}}{\sum_{k=1}^{K} e^{z_k}}$$

where z_i is the output from the final fully connected layer for class i, and K is the total number of classes.

3.4.5 Training CNNs

CNNs learn through a training process like standard neural networks. CNN training involves minimizing a suitable loss function (such as cross-entropy loss for classification) by adjusting weights and biases using gradient descent and backpropagation. Filters in convolutional layers are optimized during training to recognize task-specific patterns.

Figure 3.8 shows a basic structure of CNN architecture with 2 convolutional layer and pooling layer and a Soft Max classifier.

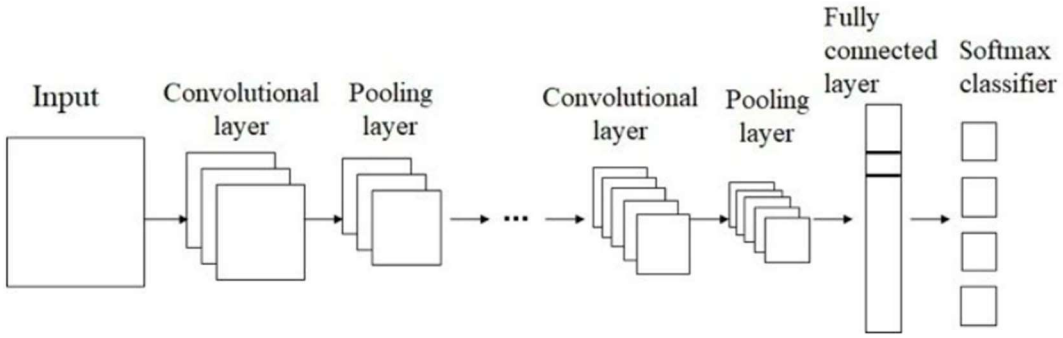


Figure 3.8: CNN Architecture (Source: Reddy et al., 2025)

3.4.6 Why CNNs are effective for images

CNNs efficiently exploit the spatial structure of images by combining:

- Local connectivity: Filters look at small regions rather than the entire image at once.
- Parameter sharing: The same filters are applied everywhere, drastically reducing the number of parameters.

- Hierarchical feature learning: Early layers learn simple features (edges, textures), and deeper layers combine these features into more complex representations (shapes, objects).

This structure allows CNNs to effectively recognize and classify images with high accuracy, making them ideal for tasks like identifying microgels in biomedical images.

3.4.7 Optimization

For this thesis, the detection network was implemented in TensorFlow/Keras and designed to operate in a YOLO-like fully convolutional manner. YOLO (You Only Look Once) is a single- stage object detection framework that divides the input image into a fixed-size grid (e.g., S×S). Each grid cell is responsible for predicting bounding boxes, objectness scores (confidence that an object exists), and class probabilities. Unlike two-stage detectors (e.g., Faster R-CNN) that first generate region proposals and then classify them, YOLO directly predicts detections in a single forward pass, enabling high-speed processing suitable for real-time applications.

3.5 Object Detection

Object detection extends image classification by identifying specific objects' locations within images. Modern object detection techniques predominantly employ CNNs in either two-stage or single-stage frameworks. Two-stage methods, such as Faster R-CNN, first generate candidate regions, subsequently classifying these regions with CNN-based classifiers (Ren et al., 2015). Single-stage methods like YOLO predict bounding boxes and class labels directly in a single CNN forward pass, significantly enhancing speed (Redmon et al., 2016). Both approaches produce bounding boxes indicating the exact position and extent of detected objects, crucial for quantifying and analyzing microgels within biological samples.

In this study, a YOLO-like single-stage detection approach was implemented. The network outputs, for each grid cell, an objectness score and bounding box coordinates, allowing direct detection of microgels in one forward pass without a separate proposal stage. This choice was motivated by the method’s computational efficiency, which is well suited for processing multiple frames from microfluidic videos in near real-time. Both objectness and bounding box predictions are essential for quantifying and analyzing microgels within biological samples. Figure 3.9 shows the different approaches used in two-stage and single-stage detection approach.

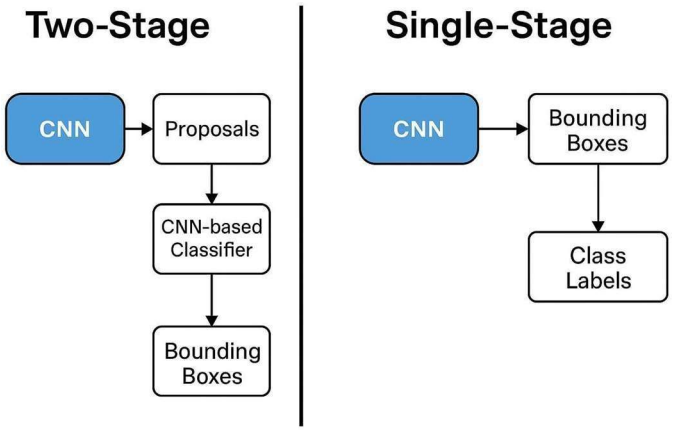


Figure 3.9: Two stage vs Single-Stage approach

3.6 Tracking and Counting Microgels Across Video Frames

The detection model outputs bounding boxes for microgels in each individual video frame. To translate these per-frame detections into a final count, a frame-tracking process was implemented. Each video is processed sequentially at a fixed frame rate (e.g., 30 FPS), and the bounding box predictions for each frame are analyzed. When a bounding box with a high objectness score (above the set detection threshold) appears in a frame, it is recorded as containing a microgel. By comparing successive frames, it is ensured that the same microgel is not counted multiple times.

This is achieved by checking whether a newly detected bounding box significantly overlaps (above an Intersection over Union threshold) with a bounding box from the previous frame. If an overlap is found, the detection is treated as the same microgel moving through the frame sequence. If no sufficient overlap is detected, it is counted as a new microgel.

This process effectively tracks microgels as they move from one side of the frame to the other, ensuring that each microgel is counted exactly once per passage through the tube. The total count for the video is then the sum of unique microgel tracks observed during processing.

Figure 3.10 shows a single microgel appearing and disappearing later.

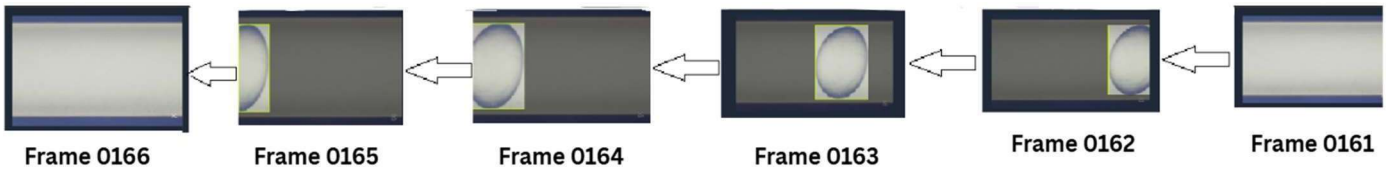


Figure 3.10: Single Microgel Across frames

3.7 **Transfer Learning** (Contextual Relevance)

CNNs typically require large datasets for effective training, often challenging in specialized biomedical applications. Transfer learning addresses this issue by reusing CNN models pre-trained on vast datasets (e.g.,

ImageNet) and fine-tuning them on smaller, specialized datasets (Yosinski et al., 2014). Initial CNN layers trained on general data capture fundamental visual features transferable to many domains, while later layers specialize through fine-tuning on the target task. This approach substantially reduces training time, computational resources, and data requirements. Transfer learning has demonstrated significant success in medical image analysis, enhancing model accuracy even when training data are limited (Shin et al., 2016).

3.8 Statistical Evaluation of CNN Predictions

The performance of CNN predictions for microgel counting is evaluated statistically using classification metrics including accuracy, precision, recall, and F1-score.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2\,x\,\frac{Precision\ x\ Recall}{Precision + Recall}$$

These metrics clearly quantify model effectiveness in accurately counting microgels within video sequences, highlighting its practical utility and reliability.

Summary

This chapter outlined the mathematical foundations and statistical methods employed within this study, encompassing image processing techniques, neural network architectures, and statistical evaluation frameworks. By clearly connecting these mathematical methods with practical implementations in detecting and counting microgels, the thesis demonstrates a cohesive approach to addressing real-world biomedical imaging challenges.

Chapter 4: Application and Case Study

4.1 Video Data Description

The study utilized three video recordings, each capturing oval-shaped like microgels moving horizontally (right to left) within a microfluidic channel. As illustrated previously in Figure 1.1 (Introduction), the experimental setup comprised a T-junction connected to PTFE tubing for the two phases, alongside a precision pump system, microscope, and computer interface for visualizing and recording droplet formation. This configuration allowed for controlled droplet generation, ensuring consistent size and flow rate for all videos in the dataset. These microgels are designed to mimic biological cells within a controlled and simplified environment to develop and test automated detection and counting methods. Each video was recorded at 30 framer per second, enabling detailed tracking of microgel movement and ensure sufficient temporal resolution for accurate analysis.

Figure 4.1 shows a video still of a microgel passing by in one of the three videos.

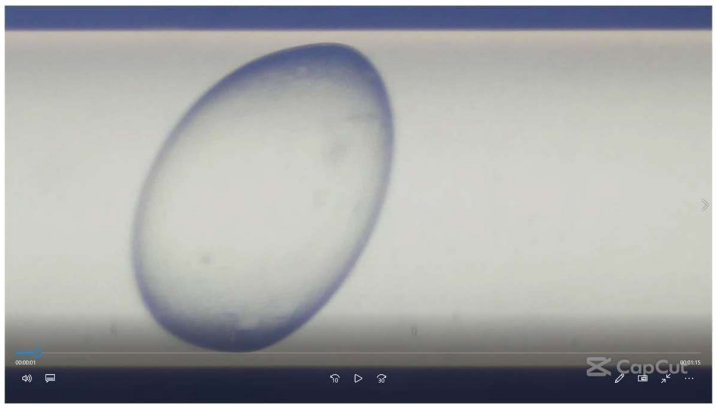


Figure 4.1: Microgel from video

4.2 Frame Extraction

To prepare the video data to training data, each frame was extracted at its original frame rate of 30 fps. This frame extraction was done in Google Colab using Python scripts which resulted in high- resolution images saved individually. This high-frequency extraction was essential to accurately capture the microgel movement across consecutive frames, facilitating precise annotation and effective training. For the second video, 1322 frames were extracted and for the third video, 2511 frames were extracted.

4.3 Manual Annotation Using Roboflow

For the first video, frames were extracted and manually annotated using Roboflow, a cloud-based image annotation tool widely adopted in image recognition projects. Roboflow allow its users to draw precise rectangular bounding boxes around the object of their interest. In this case, each individual microgel was the object of interest. These bounding boxes provide clear labels for training the CNN, specifying the exact spatial location of microgels. After annotation, Roboflow conveniently

provides the labeled dataset in the format of user's choice, YOLO for this project, this facilitates smooth integration with CNN training pipelines. For the first video 1525 frames were extracted and annotated manually as shows in figure 4.2.

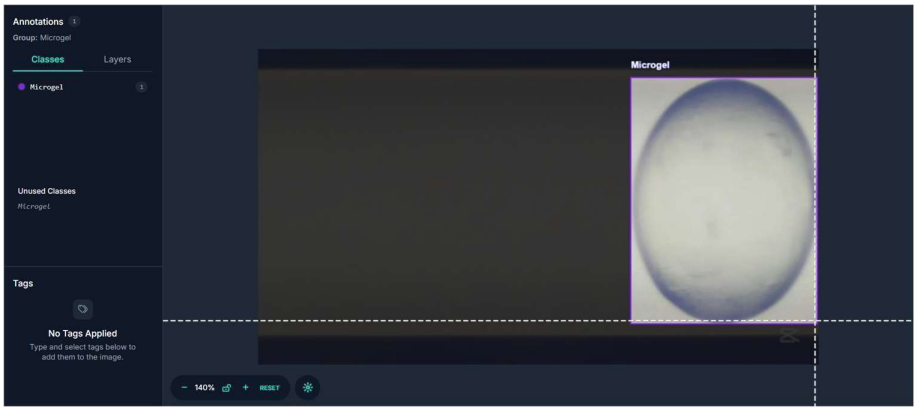


Figure 4.2: Annotating Microgel in Roboflow

4.4 Data Cleaning and Pre processing

Post-annotation, data quality was systematically enhanced using custom-written Python scripts. This was done to eliminate suspicious labeling, such as bounding box covering the entire frame, prior to feeding the dataset into the CNN model. These errors often occur due to annotation mistakes or automated labelling glitches and can mislead the model into learning irrelevant features. In our workflow, any bounding box that occupied more than a defined percentage (e.g., 90%) of the frame's width and height was automatically flagged and removed from the dataset. Figure 1.5 (Introduction) illustrated this error and an example of both correct and incorrect frames.

Frames were also categorized into different folders such as the ones containing microgels and those without microgels, creating a balanced and clean dataset. Such thorough cleaning helped to establish good CNN training and, in the end, improve the accuracy and confidence of the model.

4.5 CNN Model Training Procedure

Utilizing the carefully curated dataset, a CNN model was built and trained from scratch. The model was built so as to make the bounding boxes and through that count the microgels based on a certain threshold. This procedure involved several stages:

- Dataset Splitting: Annotated data were split into training (80%) and validation sets (20%) to monitor model performance and prevent overfitting

- CNN Architecture: A custom CNN was implemented, consisting of convolutional layers with ReLU activations, max-pooling layers for spatial dimensionality reduction, and fully connected layers for classification.

The approach in this thesis follows this principle: the model receives an RGB image input of shape (128×128×3) and progressively down samples the spatial resolution via convolutional blocks with max pooling, achieving a final grid size of 4×4. This grid size was chosen because

the backbone architecture naturally down samples the input image to a 4×4 feature map, providing an optimal trade-off between localization resolution and prediction stability.

Each convolutional block consists of:

3×3 convolution – extracts local features by sliding a learnable filter over the image or feature map.

Batch Normalization – normalizes intermediate feature maps during training to stabilize learning, reduce internal covariate shift, and allow for higher learning rates.

ReLU activation – introduces non-linearity, enabling the network to learn complex patterns.

Optional Spatial Dropout – randomly "drops" entire feature maps during training, forcing the model to avoid over-reliance on specific filters and improving generalization.

The output layer is a 1×1 convolution with five channels per grid cell, corresponding to:

Objectness score – probability that an object is present in the grid cell.

Bounding box coordinates (x, y) – relative to the grid cell's position.

Bounding box dimensions (w, h) – width and height relative to the image size.

Occasionally, two bounding boxes may be predicted for the same microgel within a single frame. This occurs when adjacent grid cells respond to different parts of the same object, often due to elongation, partial overlaps, or thresholding effects. Such duplicates are removed during post-processing by decoding all predicted boxes and applying non-maximum suppression (NMS) or local-maxima filtering. If object collisions are frequent (e.g., many microgels in proximity), the grid resolution can be increased (e.g., 6×6) to improve detection granularity, though this requires retraining to adapt to the higher resolution feature map.

These components together allow the network to predict bounding boxes and associated confidence scores for all objects in the image in a single pass, maintaining YOLO's efficiency while being adapted for our specific microgel detection task.

- Optimization and Training: The model employed the Adam optimizer and a custom loss function, optimized over 25-40 epochs based on convergence criteria

  A fully convolutional grid-based detector was trained that predicts, for each grid cell, an objectness score and four box parameters.

  Let the grid be G x G (here G =4). For image n and cell (i,j), define

$$y_{ij}^{(n)} = \left(o_{ij}^{(n)}, x_{ij}^{(n)}, y_{ij}^{(n)}, w_{ij}^{(n)}, h_{ij}^{(n)}\right) \quad , \quad \hat{y}_{ij}^{(n)} = \left(\hat{o}_{ij}^{(n)}, \hat{x}_{ij}^{(n)}, \hat{y}_{ij}^{(n)}, \hat{w}_{ij}^{(n)}, \hat{h}_{ij}^{(n)}\right)$$

Denote the ground-truth and predicted tuples as cell (i.j). The training objective combines an objectness loss (mean-squared error on ô for all cells) and a box regression loss (mean- squared error on (x̂, ŷ, ŵ, ĥ) but only for cells that truly contain an object). Formally,

$$L = \sum_{n=1}^{N} \sum_{i=1}^{G} \sum_{j=1}^{G} [objectness\ MSE\ +\ box\ MSE\ applied\ only\ when\ an\ object\ is\ present]$$

$$L = \sum_{n=1}^{N}\sum_{i=1}^{G}\sum_{j=1}^{G}\Big[(o_{ij}^{(n)} - \hat{o}_{ij}^{(n)})^2 + o_{ij}^{(n)}\big(\big(x_{ij}^{(n)} - \hat{x}_{ij}^{(n)}\big)^2 + \big(y_{ij}^{(n)} - \hat{y}_{ij}^{(n)}\big)^2 + \big(w_{ij}^{(n)} - \hat{w}_{ij}^{(n)}\big)^2 + (h_{ij}^{(n)} - \hat{h}_{ij}^{(n)})^2\big)\Big]$$

We optimize L with Adam (learning rate 10^{-3}). During inference, a cell is declared "microgel present" if $\hat{o}_{ij}^{(n)} > \tau$ (we used τ = 0.5), and its predicted bounding box ($\hat{x}$, $\hat{y}$, $\hat{w}$, $\hat{h}$) is decoded to pixel coordinates. The count for an image is the number of cells passing this threshold.

- Model Validation: Statistical metrics, such as accuracy, precision, recall and, F1 score, were monitored throughout training, ensuring robust performance evaluation

  The entire training process was conducted within Python using TensorFlow and Keras framework on Google Colab, facilitating reproducibility and accessibility.

4.6 Prediction and Microgel Counting

The trained CNN model was hence employed on frames extracted from the remaining two unseen videos. Predictions were generated on a frame-by-frame basis, which determine the presence or absence of microgel. Using additional scripts, these predictions were aggregated into a final count of microgels per video.

4.7 Applications and Practical Significance

Automated detection and counting of microgels, as demonstrated in this study, hold substantial practical relevance for biomedical and clinical applications. For example, in microfluidic flow cytometry, automated counting methods can substantially reduce processing time and human error, significantly enhancing workflow efficiency and reliability. This technique has potential utility in clinical diagnostics, especially in analyzing blood samples for circulating tumor cells (CTCs), pathogens, or other diagnostically relevant biomarkers.

Moreover, in research environments such as pharmaceutical development or cellular biology labs, precise automated counting can accelerate experimental analyses, such as measuring the effects of drugs or environmental conditions on cell viability and proliferation rates. The ability to accurately

count cellular-like entities quickly and automatically could streamline preliminary analyses, enabling researchers to rapidly screen large numbers of samples, thereby optimizing resource allocation, and reducing operational costs.

Additionally, this method serves as a valuable preliminary step toward broader automation in biomedical analysis. Future extensions could incorporate additional tasks like cell size estimation, morphological analysis, or real-time tracking of cell dynamics. Hence, the framework developed here can easily integrate into larger, more complex diagnostic and analytical pipelines, demonstrating versatility and practical adaptability for future research or clinical settings.

Summary of Experimental Process

This chapter provided a comprehensive overview of the applied methodology, from data preparation through model training and application to unseen videos. The case study demonstrates the practical viability and potential impact of integrating computational methods within biomedical imaging workflows, emphasizing simplicity, accuracy, and reliability. The developed pipeline thus represents a foundational step towards more extensive automation in biomedical diagnostics and research.

Chapter 5: Results

This chapter presents the outcome of applying the CNN model trained in Chapter 4 to detect and count microgels in two unseen microscopy videos. The results are evaluated in terms of accuracy, precision, recall and F1 score to comprehensively measure the model's effectiveness and reliability.

5.1. Model Training

The model was trained for 25 epochs with a batch size of 8, and training/validation loss was monitored throughout. Figure 5.1 shows the evolution of the training and validation loss over epochs. The steep initial drop in both curves indicates rapid convergence in early epochs, followed by gradual stabilization. The minimal gap between training and validation curves suggests limited overfitting, indicating good generalization to unseen data.

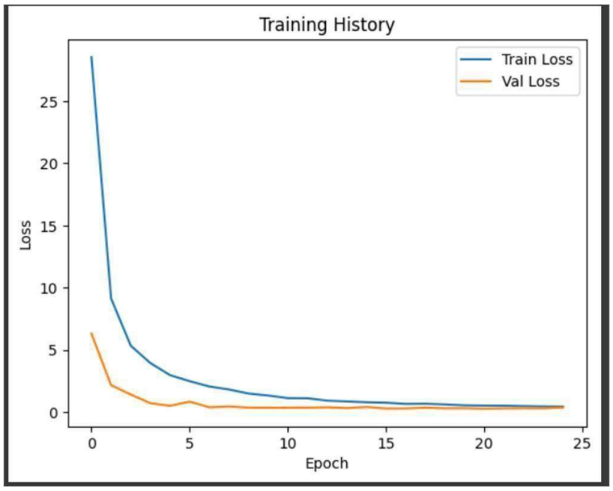


Figure 5.1: Training and validation loss history for the proposed CNN-based microgel detection model.

5.2. Model Performance on Validation Data

To assess the generalization ability of the trained CNN, the model was evaluated on a validation dataset from video 1. Figure 5.2 illustrates an example prediction, showing the bounding boxes generated by Roboflow and the model for detected microgels. The example demonstrates that the model can accurately localize and count microgels under varied flow patterns and lighting conditions present in the validation set.

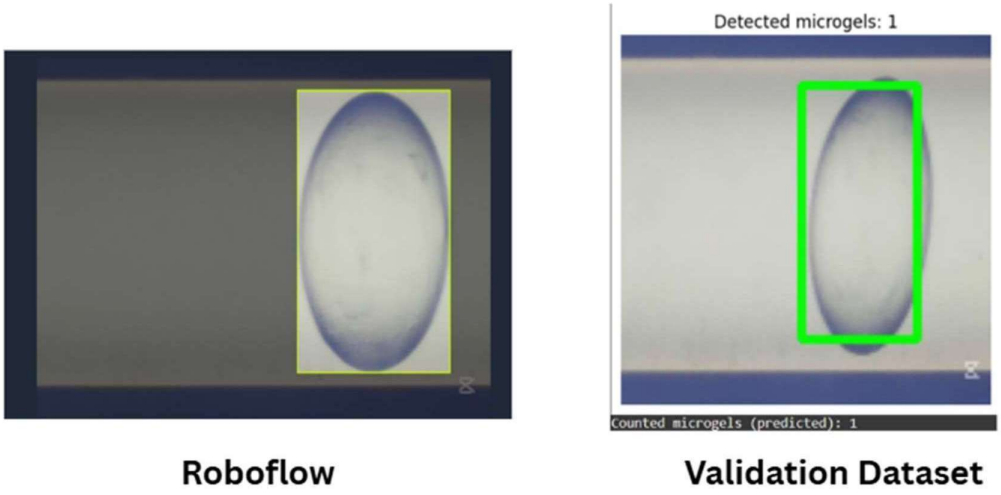


Figure 5.2: Example shows a) Bounding Box made by Roboflow b) Bounding Box generated through Validation Dataset

The predictions closely match the ground truth annotations, supporting the reliability of the trained model in processing frames outside the training set.

5.3. Evaluation Metrics

For better understanding the model's performance, standard statistical evaluation metrics were computed. These metrics were based on the prediction of the bounding boxes:

- Accuracy: Measures overall correctness across all frames
- Precision: Reflects the proportion of frames correctly predicted to contain microgels
- Recall: Indicates how effectively the model identifies frames containing microgels
- F1-score: Balances precision and recall, providing a unified measure of performance

Precision	Recall	F1 Score
0.959	0.959	0.959

Table 5.1: Statistical Evaluation Metrics for CNN Microgel Detection

These evaluation metrics confirm that the CNN model achieves high accuracy and reliability in predicting microgel presence across frames.

5.4. Visualizing Model Performance

Results

To visually interpret the CNN model's effectiveness, some video frames are presented in Figure 5.3, 5.4 and 5.5, illustrating instances of correct detections (true positives), incorrect detections (false positives), and missed detections (false negatives).

- True Positives: Correctly identified microgels
- False Positives: Incorrectly identified non-existent microgels
- False Negatives: Missed microgels

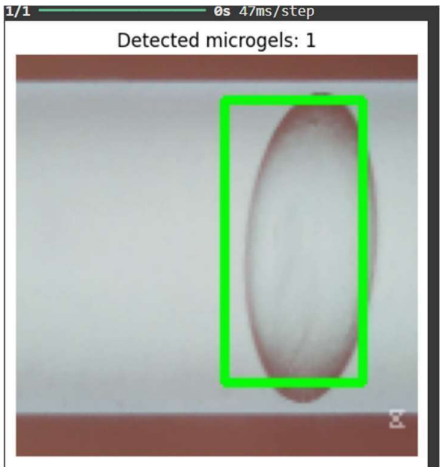


Figure 5.3: Correctly predicted microgel

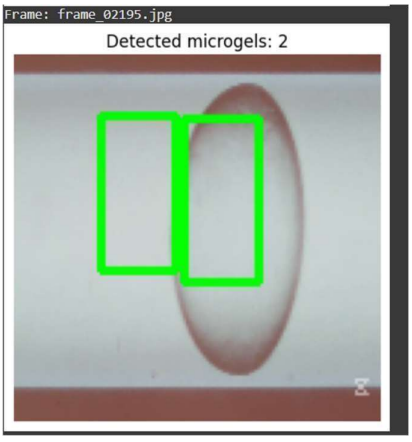


Figure 5.4: Incorrectly predicted microgel (One is correctly predicted)

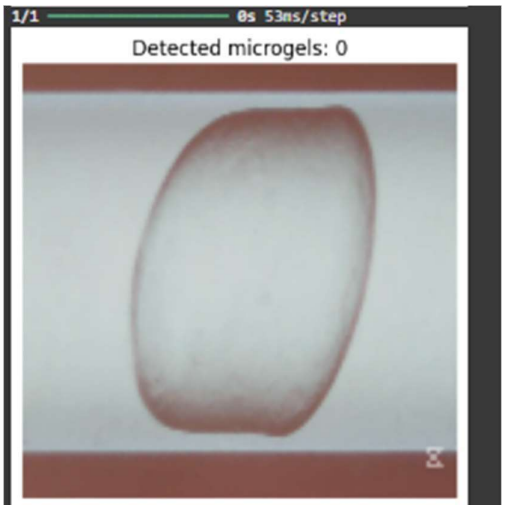


Figure 5.5: Missed Microgel

5.5. Microgel Detection Results

The frames obtained from the second and third video sequences were only used for prediction. The CNN model was first trained exclusively on frames extracted from the first video sequence, and then applied to the second and third sequences without any additional retraining. This ensured that the model's performance on videos 2 and 3 reflected its ability to generalize to unseen data. Individual predictions were combined to yield a total microgel count of each video sequence. The table 5.2 summarizes the counting results obtained by the CNN model.

Video Sequence	Total Frames	Microgels Detected
Video 2	1322	336
Video 3	2511	645

Table 5.2: Microgel Count results from CNN model

5.6. Discussion of Results

In this study, the findings show that the Convolutional Neural Network model can detect and counting microgels in microscopy videos. The model predictions and manual verification percentage agreements were always high than 97 percent which implies that the model was robust and very reliable to be applied to practice.

However, the model did exhibit some minor errors. Upon inspection, the bounding box were not covering the microgels entirely, some partially visible microgels near frame edges were missed with some variations in lighting conditions, or minor annotation inaccuracies in the training set. Such issues highlight the importance of careful data annotation and potentially adopting more sophisticated preprocessing steps in future studies.

Importantly, the lightweight nature of this CNN model was trained entirely from scratch with limited data and this emphasizes practicality and demonstrates that complex pre-trained models or substantial computational resources are not strictly necessary to achieve reliable detection performance in specific biomedical contexts.

5.7. Practical Implications of the Results

The robust performance of this model has a direct application relevance for clinical and laboratory settings, where accurate and rapid detection and counting of cellular structures can significantly reduce manual labor and improve throughput. Potential applications extend to field such as cancer diagnostics, pathogen detection, and drug screening assays, where cell or particle enumeration is critical.

Furthermore, this simple yet accurate enough approach could easily integrate into broader analytical pipelines which provides initial screening or triaging to prioritize samples requiring detailed human expert analysis. By automating routine counting tasks, clinical laborites can achieve improved diagnostic workflows, faster turnaround times, and reduced human error.

Summary

This chapter has broadly ascertained that the convolutional neural network (CNN) paradigm is able to detect and count microgels with an excellent degree of consistency and accuracy throughout the microscopy video streams. The results support the effectiveness of the systematic yet simple methodological design, used in the thesis. The study thus supports the future applicability and the feasibility of CNN-based object detection to clinics and in research in the field of biomedicine.

Chapter 6: Discussion and Conclusions

6.1. Reflection on Project Goals

In the current study, the aim was to develop an algorithmic framework that is lightweight, interpretable, and robust enough to detect and count microgels in the microscopy videos, which is equivalent to automated cell counting in biomedical diagnostics. The appropriate study accordingly took into consideration the well-tuned flow of basic image-processing operations, rigorous annotation, and a scratch-trained custom convolutional neural network (CNN). The obtained model had an excellent accuracy and showed a good generalization to unknown videos, thus confirming the effectiveness of the pursued solution on practicality.

As shown through the findings, even in such a pretext of having to deal with a low amount of data and computational facilities it is quite possible to deliver a robust performance by keeping in mind the quality of a dataset, the specificity of tasks at hand and its clarity using mathematical principles rather than only depending upon the pre-existing training architectures or large training datasets. The application of rigorous principles of mathematics to data representation, training protocols, and the measurement of performance provides a concise example of how object-detection methodologies can be adapted to a biomedical setting upon which the project is founded.

6.2. Strengths of the Approach

The reason behind its high efficacy displayed in this work can be seen in the simplification of the work. The model succeeded in internalizing meaningful spatial patterns by avoiding architectural selections that are complicated and using manual annotations of high quality. Besides that, the modular and transparent construction of the end-to-end architecture, i.e., involved in the frame extraction, detection, and counted, was deliberately avoided, thus making it both explainable and replicable.

Moreover, the fact that the project is based on the frame-level supervision and binary detection logic. This form of implementation and validation is thus interesting in terms of requirements, since implementation and validation work well, and it particularly fits well in clinical and research conditions with time and computational limitations.

Also, the integration of strict data-cleaning scripts, such as elimination of frames with invalid bounding boxes, reduced the effects of annotation noise, which allowed increasing the stability of models. The use of Google Colab and Roboflow also promotes access and usability, which makes the utilization of similar pipelines by investigators working in low-resource backgrounds or during initial experimentation more likely to be given consideration.

6.3. Limitations

Despite its success, the project has a few limitations. This trained convolutional neural network (CNN) was tested using a limited collection of videos having a common environment which was homogeneous. As a result, the applicability of the model to generalize on differing experimental design, lighting conditions, or morphologically diverse microgels has not been determined. In addition to this, the pipeline does not carry out diameter or shape-based measurements even

though its priorities were based on detection and counting. This limitation restricts the level of analysis of the model especially in the cases where an application requires the examination of morphology.

Contemporary such solutions offer frame-wise detection, but no specific temporal tracking of microgels; in other words, no algorithm to track microgels through successive frames. The resulting lack of temporal coordination can lead to some undercounting and overcounting: there is the chance that microgels splitting within a frame will be counted many times, and split pieces that fall between frames may go uncounted altogether.

Moreover, due to computational deficient system, uploading of data took quite a long time than expected. A better device with better computational features would be advised.

Adding to this, in testing the data the bounding boxes did not quite fit the microgels perfectly. Training the data more would overfit the data. It is possible that using a pre-trained model would have avoided this.

Lastly, manual annotation was used to create the ground truth of the methods of supervised learning. This traditional procedure does however add subjective bias into the process as well as increasing possibility of human error, particularly at edge cases where the microgels are not fully visible.

6.4. Future Work and Extensions

Several promising directions can be explored for future research:

- Temporal Modeling: The inclusion of sequential models, such as recurrent neural networks or optical flow methodologies, could enhance microgel tracking across video frames. Prior research has successfully applied RNNs to video object detection (Qasim et al., 2020), and recurrent networks with motion-aware attention have shown robustness in spatial-temporal tracking (Fan et al., 2021).

- Shape and size estimation: Beyond detection, measuring microgel dimensions would deepen analysis. For instance, microgel diameters can be accurately measured using tools like ImageJ from microscopy videos (Qazi et al., 2022).

- Transfer learning or pre-trained models: Leveraging pre-trained models such as ResNet or VGG through transfer learning may improve detection accuracy and reduce training time. This approach has proven effective in object detection domains (Karpathy et al., 2015).

- Edge Detection: Methods such as the Canny or Sobel operator identify object boundaries (Marr and Hildreth, 1980). Could be used for microgel shape estimation.

- Background Subtraction: Useful in isolating moving objects in video sequences (Stauffer and Grimson, 1999). Could reduce false detections in static backgrounds.

- Classic Image Processing: Introducing classical techniques such as contrast normalization, adaptive thresholding, and morphological operations like opening and closing, could better prepare images for downstream detection stages (Soille et al., 1999).

- Annotation automation: To reduce manual annotation workload, semi-automated and AI- assisted labeling strategies—employing active learning or weak supervision—can speed up data preparation while maintaining quality (Alamu et al., 2023, Sutharsan et al., 2023).

- Generalization: Evaluating the model's performance on data with variable experimental conditions, such as different channel geometries or illumination patterns, would validate its robustness. Deep learning models have successfully generalized across different microscopy modalities (Krikid et al., 2024).

6.5. Broader Implications

This work contributes to a growing body of research that seeks to simplify and democratize the use of deep learning in biomedical image analysis. By showing that even a modest CNN can deliver reliable counting in real biological scenarios, the project encourages researchers to focus on task design, data quality, and mathematical modeling over complexity for its own sake.

The practical relevance is notable in microfluidic systems or flow cytometry-based diagnostics, automated counting can significantly reduce manual workloads and increase throughput. Such techniques may be especially valuable in point-of-care testing environments or resource-limited settings.

Conclusion

In conclusion, this thesis presents a compact, accurate, and interpretable pipeline for microgel detection and counting using a custom-trained CNN model. By grounding the work in strong mathematical foundations and focusing on practical usability, the project successfully bridges the gap between theoretical modeling and experimental application. While there is room for future expansion in terms of functionality and generalization, the current framework serves as a robust foundation for further research into automated image and video analysis in biomedical contexts.

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Appendices

Appendix A — Code Listings

- **Part_1_Roboflow_code.ipynb**:
  Prepares training dataset using manual annotations, filters incorrectly labelled images, and separates frames based on microgel presence.

- **Part_2_Extracting_frames.ipynb**:
  Extracts frames from video at 30 fps, saving them into structured folders for annotation and prediction.

- **Part_3_Model_Buidling.ipynb**:
  Defines and trains a CNN from scratch using the cleaned dataset. Includes model training, validation, and prediction on unseen data.

Appendix B — Selected Screenshots

- Sample annotated frame from Roboflow

- Example of false positive and true positive detections

- Sample frame showing bounding boxes correctly placed

Appendix C — Additional Mathematical Details

- Derivation of binary cross-entropy loss

- Convolution operation walkthrough with example

- CNN architecture layer-by-layer breakdown (diagram/table format)